# Homework 2 CS 555 Spring 26

Md Istiaque Ahmed (miahmed2)

March 13, 2026

## Question 1: 1D Burgers Equation

In this problem, we numerically solve the 1D Burgers equation:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \tag{1}$$

on the domain $\Omega = [0, 1]$ with homogeneous Dirichlet boundary conditions and initial condition $u^0(x) = \sin(\pi x)$. The viscosity is set to $\nu = 1/(100\pi)$.

### Part (a): Solution Snapshots

The equation is integrated using finite differences with 3-point stencils for spatial derivatives and a 3rd-order Implicit-Explicit (IMEX) scheme in time (BDF3 for the viscous term, EXT3 for the convective term). The time-stepping is bootstrapped using 1st and 2nd-order methods for the initial two steps.

Figure 1 displays the computed solution profile $u(x, t)$ at intervals of 0.1 up to $t = 2.0$. As expected, the initial sine wave advects to the right and steepens, forming a sharp boundary layer against the wall at $x = 1$.

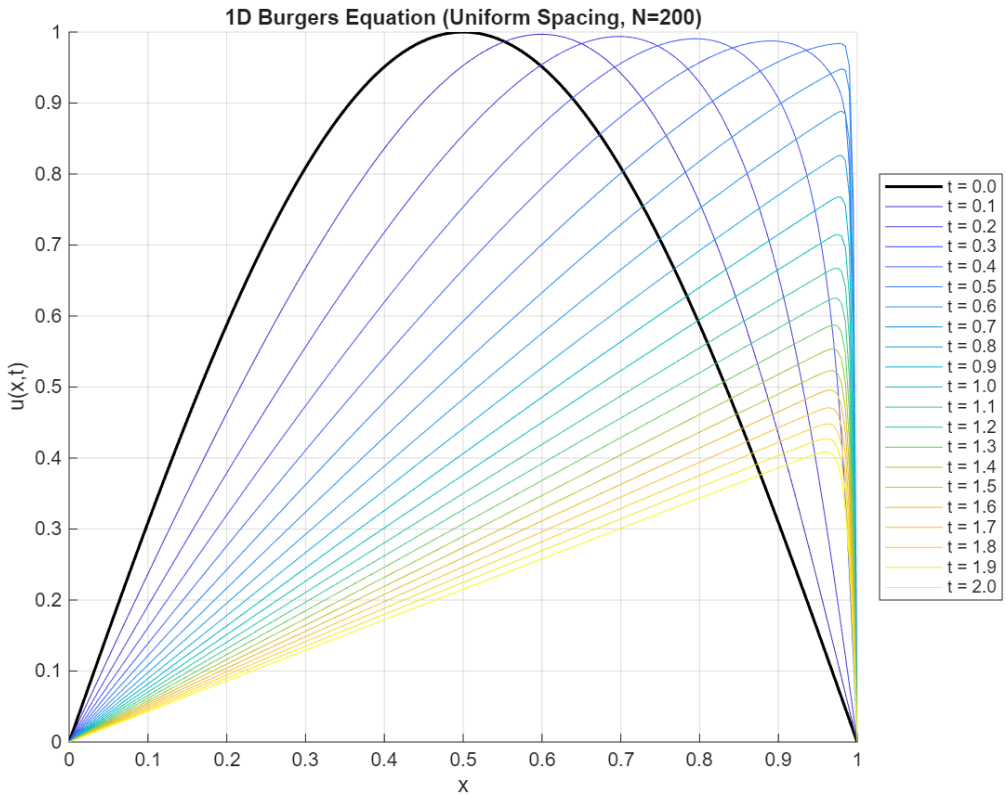


Figure 1: Evolution of the 1D Burgers equation on a uniform grid ($N = 200$). Snapshots are taken every 0.1 seconds.

## Part (b): Maximum Spatial Derivative

We compute $s(t) = \max |\partial u/\partial x|$ over the domain for $t \in [0, 2]$. The time step is chosen to satisfy CFL $= 0.1$. Assuming a maximum wave speed of $|c| \approx 1.0$ and uniform spacing $\Delta x = 1/200$, this yields a stable timestep of $\Delta t = 0.0005$. The computed parameters for the uniform grid ($N = 200$) are:

- **Maximum slope:** $s^* = 208.77638$
- **Time of maximum:** $t^* = 0.51400$

Figure 2 visualizes the evolution of this derivative. The qualitative behavior agrees with Figure 6 in the Basdevant et al. reference paper, exhibiting a rapid spike to a maximum value followed by a gradual viscous decay. Because Figure 6 is plotted against $\pi t$, our peak occurring at $t \approx 0.51$ correctly aligns with their reported peak near $\pi t \approx 1.6$.

Comparing our numerical values to Table 3 of the reference (analytical $s^* \approx 152.005$ at $t^* \approx 0.51$), the temporal location $t^*$ is highly accurate. However, the computed magnitude $s^*$ exhibits a significant overshoot (208.78 vs. 152.00). This discrepancy is a direct consequence of numerical dispersion; the uniform spacing of $\Delta x = 1/200$ is too coarse to adequately resolve the boundary layer, introducing non-physical, spurious oscillations that artificially inflate the maximum derivative.

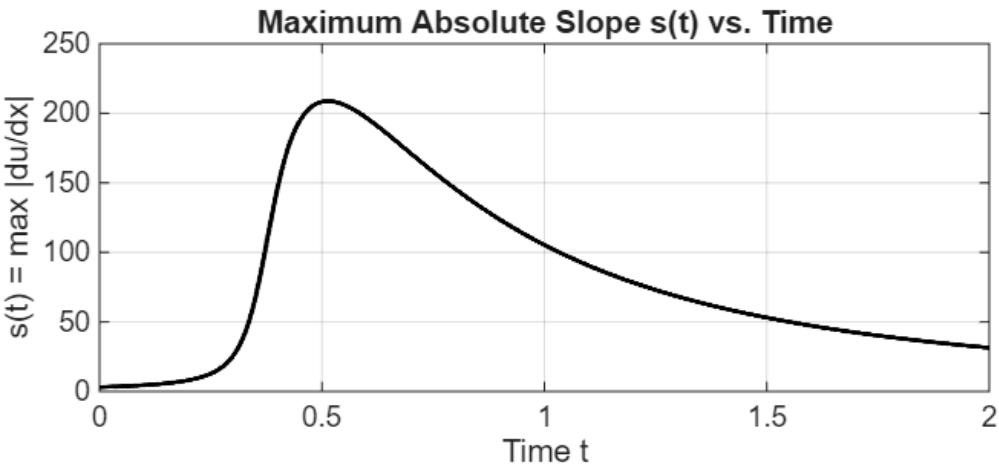


Figure 2: Maximum absolute spatial derivative $s(t)$ versus time.

## Part (c): Convergence with Uniform Spacing

We compute $s^*$ as a function of $N = 2^k$ for $k = 5, 6, \ldots, 11$ using uniform spacing and the convective formulation. The tabulated relative errors against the analytical maximum $s^* = 152.00516$ are presented in Table 1.

Table 1: Convergence of $s^*$ for Uniform Spacing (Convective Form)

| $N$ | nsteps | $s^*$ | rel.err. |
|---|---|---|---|
| 32 | 640 | 204.75163 | $3.47004 \times 10^{-1}$ |
| 64 | 1280 | 264.95618 | $7.43074 \times 10^{-1}$ |
| 128 | 2560 | 241.56922 | $5.89217 \times 10^{-1}$ |
| 256 | 5120 | 192.58263 | $2.66948 \times 10^{-1}$ |
| 512 | 10240 | 164.69320 | $8.34711 \times 10^{-2}$ |
| 1024 | 20480 | 155.38910 | $2.22620 \times 10^{-2}$ |
| 2048 | 40960 | 152.86551 | $5.66003 \times 10^{-3}$ |

## Part (d): Conservation Form at Finest Resolution

The nonlinear advection term can be recast into the conservation form: $\frac{1}{2}\frac{\partial u^2}{\partial x}$. Testing this approach at the finest uniform resolution ($N = 2048$, $\Delta t \approx 4.88 \times 10^{-5}$) yields:

- **Maximum slope:** $s^* = 152.42586$
- **Time of maximum:** $t^* = 0.51050$

While the physical formulations are analytically equivalent, the discrete truncation error differs from the convective form, resulting in a slightly modified approximation of the maximum slope.

## Part (e): Convergence Study and Comparison

The following tables document the convergence behavior for $N = 2^k$ ($k = 5, \ldots, 11$) across four combinations of spatial discretization and nonlinear formulation. As instructed, the timestep $\Delta t =$

$0.1/N$, derived from the uniform grid CFL condition, is held constant across all cases to ensure a direct comparison without introducing variable time-stepping artifacts. The convergence ratio is defined as $\mathrm{err}_{N/2}/\mathrm{err}_N$.

Table 2: Case i. Uniform spacing, convective form

| $N$ | nsteps | $s^*$ | rel.err. | ratio |
|---|---|---|---|---|
| 32 | 640 | 204.75163 | $3.47004 \times 10^{-1}$ | - |
| 64 | 1280 | 264.95618 | $7.43074 \times 10^{-1}$ | 0.467 |
| 128 | 2560 | 241.56922 | $5.89217 \times 10^{-1}$ | 1.261 |
| 256 | 5120 | 192.58263 | $2.66948 \times 10^{-1}$ | 2.207 |
| 512 | 10240 | 164.69320 | $8.34711 \times 10^{-2}$ | 3.198 |
| 1024 | 20480 | 155.38910 | $2.22620 \times 10^{-2}$ | 3.749 |
| 2048 | 40960 | 152.86551 | $5.66003 \times 10^{-3}$ | 3.933 |

Table 3: Case ii. Chebyshev spacing, convective form

| $N$ | nsteps | $s^*$ | rel.err. | ratio |
|---|---|---|---|---|
| 32 | 640 | 167.45688 | $1.01653 \times 10^{-1}$ | - |
| 64 | 1280 | 153.42089 | $9.31369 \times 10^{-3}$ | 10.914 |
| 128 | 2560 | 152.11972 | $7.53643 \times 10^{-4}$ | 12.358 |
| 256 | 5120 | 152.01843 | $8.72962 \times 10^{-5}$ | 8.633 |
| 512 | 10240 | 152.00751 | $1.54404 \times 10^{-5}$ | 5.654 |
| 1024 | 20480 | 152.00569 | $3.48080 \times 10^{-6}$ | 4.436 |
| 2048 | 40960 | 152.00529 | $8.50164 \times 10^{-7}$ | 4.094 |

Table 4: Case iii. Uniform spacing, conservation form

| $N$ | nsteps | $s^*$ | rel.err. | ratio |
|---|---|---|---|---|
| 32 | 640 | 65.52165 | $5.68951 \times 10^{-1}$ | - |
| 64 | 1280 | 102.90704 | $3.23003 \times 10^{-1}$ | 1.761 |
| 128 | 2560 | 140.71434 | $7.42792 \times 10^{-2}$ | 4.348 |
| 256 | 5120 | 157.73907 | $3.77218 \times 10^{-2}$ | 1.969 |
| 512 | 10240 | 156.53456 | $2.97977 \times 10^{-2}$ | 1.266 |
| 1024 | 20480 | 153.55362 | $1.01869 \times 10^{-2}$ | 2.925 |
| 2048 | 40960 | 152.42586 | $2.76767 \times 10^{-3}$ | 3.681 |

Table 5: Case iv. Chebyshev spacing, conservation form

| $N$ | nsteps | $s^*$ | rel.err. | ratio |
|---|---|---|---|---|
| 32 | 640 | 159.19227 | $4.72820 \times 10^{-2}$ | - |
| 64 | 1280 | 153.81930 | $1.19348 \times 10^{-2}$ | 3.962 |
| 128 | 2560 | 152.15108 | $9.59988 \times 10^{-4}$ | 12.432 |
| 256 | 5120 | 152.01876 | $8.94915 \times 10^{-5}$ | 10.727 |
| 512 | 10240 | 152.00711 | $1.28268 \times 10^{-5}$ | 6.977 |
| 1024 | 20480 | 152.00556 | $2.62221 \times 10^{-6}$ | 4.892 |
| 2048 | 40960 | 152.00525 | $6.23135 \times 10^{-7}$ | 4.208 |

**Discussion**

- **Grid Spacing (Uniform vs. Chebyshev):** The uniform grid severely under-resolves the sharp internal layer at lower $N$ values, leading to significant numerical dispersion. This manifests as poor approximations of $s^*$ and erratic error convergence until $N \geq 1024$. In contrast, Chebyshev spacing naturally clusters grid points at the boundaries, placing high resolution precisely where the local gradients are maximized. Consequently, the Chebyshev cases reach an $\approx 10^{-5}$ relative error by just $N = 256$, outperforming the uniform grid's finest resolution.

- **Convergence Ratio:** The theoretical error ratio for a second-order scheme when halving the spatial step size is $\approx 4$. For the Chebyshev grids (Cases ii and iv), the ratio rapidly stabilizes and approaches $\approx 4.1$ by the finest resolution, strictly confirming the scheme's second-order spatial accuracy. For uniform grids, the ratio behaves erratically before finally approaching the theoretical asymptotic value once the grid properly resolves the boundary layer at high $N$.

- **Formulation (Convective vs. Conservation):** At lower resolutions, the discrete truncation errors of the two forms lead to distinct failure modes; the convective form overestimates $s^*$ due to spurious oscillations, while the conservation form under-predicts it. However, once the grid is sufficiently refined (especially with Chebyshev spacing), both formulations converge to the analytical maximum $s^* \approx 152.005$ with nearly identical precision, robustly validating their formal equivalence.